Deep Space Relay Resource Planning Model Based on Sliding Time Window

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Abstract—When a deep space probe enters into the back of the celestial body, it cannot directly communicate with the earth for TT&C (Telemetry, Tracking & Control) and data transmission. As a transfer station for communication between the earth TT&C station and the deep space probe, the deep space satellite realizes all-weather data transmission between the earth and deep space probe. Because of the exclusivity of the deep space relay satellite communication link, several deep space probes compete for relay resources. Aiming at the problem of relay resource planning for deep space probes, this paper proposes a deep space relay resource planning algorithm based on sliding time window, designs hierarchical planning and scheduling architecture of relay resources, and realizes automatic planning and allocation of relay resources and unified scheduling and management. The practical application results show that the algorithm can meet the requirements of resource planning for celestial backside exploration tasks, improve the efficiency of resource allocation and reduce the rate of resource fragmentation.

Keywords—Celestial backside detection, Deep space relay satellite, Resource planning, Time slide windows

I. INTRODUCTION

Due to the space shielding of the celestial body, the space probe cannot communicate directly with the ground TT&C station when entering the back of the target celestial body. So it is necessary to provide the communication support between the ground TT&C station and the space probe through the deep space relay satellite when arriving at the back of the target celestial body, which is the prerequisite for achieving the mission of detecting the back of the target celestial body[1]. When multiple detectors work independently or cooperatively on the back side of the target celestial body, the ground control center needs to send corresponding control commands or inject data through the uplink channel of Deep space relay satellite. As the uplink channel of Deep space relay satellite is unique, the working mode of time-sharing uplink is usually adopted between Deep space relay satellite and multiple detectors. In the control process of complex operation with multiple detectors cooperating with each other, the ground control center needs to fully consider the coupling, time and resource constraints between multiple detectors to complete the cooperative control based on Deep space relay satellites. In this paper, a multi-layer scheduler and a relay resource planning algorithm are designed to automatically allocate relay TT&C resources for Deep space exploration, so as to realize optimal scheduling and conflict resolution for different detector exploration tasks.

II. FEATURES OF CELESTIAL BACKSIDE DETECTION MISSION

Since the probe on the back of the celestial body cannot directly communicate with the TT&C station, relay communication becomes the first problem to be solved in the detection of the back of the celestial body. The deep-space relay satellite runs in the corresponding relay orbit, and provides necessary measurement and control communication support for the probe on the back of the celestial body [2]. Deep-space relay satellites provide forward/reverse real-time and delayed relay communication between multiple probes on the back of celestial bodies and TT&C stations. As shown in Fig.1, the characteristics of deep-space relay satellite communications include [3,6]:

1) The control of the probe is realized through the serial communication link of the TT&C station and the deep-space relay satellite. The communication of the deep-space relay satellite is different from the ordinary ground station, and it is not simply a command and data transmission path. The flight orbits of deep-space relay satellites are constantly changing, the relative positions and distances of the earth and the satellite are also changing, and the available communication period is also changing. Deep-space relay satellites require periodic orbit maintenance. During the orbit maintenance period, the communication with the probes on the back of the celestial body cannot be guaranteed.

2) The deep-space relay satellite can use the USB TT&C system for the ground measurement and control system. The relay communication link to the probe is X-band, and the forward code rate is relatively low, the data transmission rate

can be switched on demand, and the transmission time is relatively long and variable.

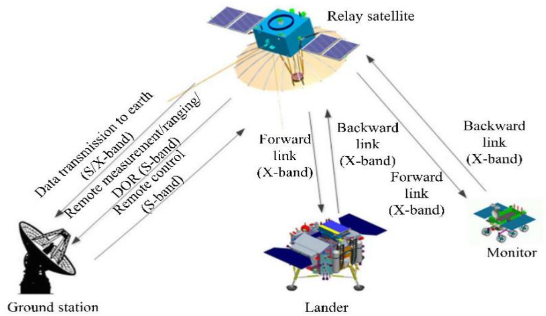

Fig. 1. Schematic diagram of multi-probes communication based on Deep Space Relay Satellite

Combining the characteristics of deep-space relay satellites, analyzing the measurement and control characteristics of different probes on the back of celestial bodies, sorting out and summarizing the probe's requirements for relay resources, it mainly includes the following aspects:

- Complex resource constraint requirements for deep-space exploration. Due to the difference of the exploration, space probes have different requirements for relay resources, and there are also significant differences in the discrete and continuous requirements of relay resources.
- The environment of the detection on the back of the celestial body is unknown and the situation is more complicated, making it difficult to predict the state of the probe. When an emergency situation occurs in an exploration, the ground control center must not only ensure the reliability of normal control, but also have the ability to handle emergency response. Therefore, it is necessary to realize the preemption and reallocation of relay resources, and realize the rapid intervention of subsequent tasks.
- Due to the long duration of deep-space exploration missions, a variety of exploration missions should be arranged in a centralized and continuous relay satellite measurement and control arc, so as to maximize the number of missions and implementation efficiency and reduce the pressure of the staff on the ground.
- When assigning relay resources, it is necessary to consider the mutual influence and constraint relationship between different tasks. For example: the previous tasks can affect the planning of subsequent tasks, but the subsequent tasks cannot affect the previous one.

In view of the complex time constraints, resource constraints, concurrency and other characteristics of deep-space explorations, relay resource planning not only needs to consider task selection and sequencing, but also rationally allocate resources and time. It is quite different from the existing task scheduling algorithm[7-10]. The paper [11] proposes a minimum data loss scheduling algorithm for earth-moon relay data resource allocation and scheduling problem, which focuses on delay-tolerant task scheduling and data transmission efficiency and fails to meet the needs of real-time measurement and control. This paper proposes a deep-space exploration relay resource planning algorithm based on sliding time window. Sufficient consideration of above-mentioned characteristics, it comprehensively considers different requirements, used relay resources and remaining relay resources, and establishes a resource planning model. The model accomplishes the application, acceptance, conflict detection and resolution of the relay resources. It effectively solves the contradiction between the supply and demand of limited deep-space relay resources in explorations, and provides technical support for optimizing the relay resources planning.

III. Hierarchical Planning And Scheduling Of Relay Resources

To meet the requirements of explorations for celestial backside detection, relay resources need to support the platform maintenance, orbit maneuver, load control and patrol detection of multiple detectors and other measurement and control digital tasks. The mission of celestial backside detection comes from different users, and the task priority and measurement and control characteristics are also different. According to the needs, different data transmission modes of real-time and delay are adopted, so there are great differences in the demand for relay resources [12].

The ground control center classifies different detection tasks according to measurement and control characteristics, and designs perfect resource allocation principles and requirements in advance, which are defined as a standardized allocation strategy. Different tasks of measurement and control are interrelated with each other and need to be extracted into a unified constraint rule library. This paper designs the planning and scheduling method and system structure of deep-space relay resources. The relay resource hierarchical planning and scheduling system includes: global scheduler, local scheduler, constraint rule library, allocation strategy library, and distributed information management system. The global scheduler is responsible for the overall planning of multi-probe tasks, and the local scheduler implements single-detector relay resource allocation. The distributed information management system is responsible for the release of relay resources, realizes the acceptance and summary of multi-probe, multi-user applications. It also displays the results of relay resource planning to users in real time, to support the decision-making of subsequent tasks.

According to the requirements for relay resources in the detection task on the back of the celestial body, the method of hierarchical planning of relay resources is designed to complete the process of multi-user demand acceptance, resource management, and hierarchical planning based on constraint rules and allocation strategies. The relay available window is the basis for the planning and allocation of relay resources. Based on the deep-space relay satellite TT&C arc, the ground control center calculates some TT&C arcs, which can be used for data transmission, that is, the available relay windows.

The available relay windows include data elements such as start time, end time, tracking time, TT&C station and TT&C

type. Requirements of relay resource are proposed by different users such as probe platform management, load, patrol exploration, etc. It is the input condition for relay resource planning and allocation. Task requirements mainly include: objects, duration requirements, time constraints and key level information. The task of deep-space exploration relay resource planning and scheduling is to allocate the available relay window according to the relay resource requirements submitted by users, and to schedule and execute different tasks to maximize the utilization rate of relay resources. The implementation of relay resource hierarchical planning and scheduling is as follows:

1) In the preparation stage of the relay resource planning, the ground control center calculates the available relay resources according to the visible arc of the deep-space relay satellite, and outputs available relay windows.
2) Release available relay windows through distributed information management system for users of different objects. Based on the control requirements of the detector, each user proposes the requirements of relay resources. The system automatically completes the data collection and carding, and submits to the global scheduler.
3) The global scheduler preprocesses the requirements of relay resources, loads the information of the constraint rule library and allocation strategy, deals with the needs of each user, and realizes multi-objects resource planning. The output is the available relay resources of a single object, which is the input of corresponding local scheduler.
4) The local scheduler calls the relay resource allocation algorithm to plan and allocate the user requirements of a single object, calculate and resolve resource conflicts, and output the relay resource allocation result and rejected list of tasks. The distributed information management system receives and confirms the relay resource allocation results, and performs detection tasks according to the plan.

IV. Resource Allocation Model Based On Sliding Time Window

Suppose the task set $T = \{t_1,\cdots,t_n\}$, any task $t_i \epsilon T$ can be expressed as a multivariate group (p_i,C_i,d_i). Here p_i is the priority of t_i. C_i is time constraint window of t_i, which can be expressed as $[c_{ib},c_{ie}]$,where c_{ib} is the earliest start time of t_i,and c_{ie} is the latest end time of t_i. d_i is the interval time of t_i.

The set of relay visible windows is denoted as $W = \{w_{1,}\cdots,w_k\}$. Any window $w_j \epsilon W$ can be expressed as $[w_{jb}, w_{je}]$. w_{jb} is the start time of the relay visual window, and w_{je} is the end time of the relay visual window.

The relay window assigned to t_i is $R_i = \{r_{i1},r_{i2}...r_{im}\}$. r_{ij} can be expressed as$[b_{ij}, e_{ij}]$, where b_{ij} represents the start time of the relay window and e_{ij} represents the end time of the relay window.

1) Distribution constraints

According to the characteristics of the detection of the back of the celestial body, the task needs to meet the following constraints:

a. The constraint of Visual window. Only when the relay satellite is in the ground TT&C arc, the probe can communicate with the ground through the relay satellite. Therefore, the task must be executed within the visible time window of the relay satellite, i.e.

$$\forall r_{ij},\ r_{ij} \subseteq \bigcup_{j=1}^{k} w_j \tag{1}$$

b. The constraint of observation time. Some task must finish on time strictly, which are mainly reflected in the earliest start time and end time, i.e.

$$\forall r_{ij},\ b_i \geq c_{ib}, e_i \leq c_{ie} \tag{2}$$

c. Non-overlapping constraints. Relay resources are exclusive. Only one target can be observed at any time and can only be assigned to one task at a time. Therefore, the relay resources allocated to any two tasks cannot overlap, and the following constraints should be met:

$$\forall t_i,t_j \in T, R_i \cap R_j = \emptyset \tag{3}$$

d. Concentration constraint: In order to reduce fragmentation and improve utilization of relay resources, adjacency tasks can be allocated to a continuous relay visual window as much as possible.
e. Satisfaction constraints. Since emergency operations which have higher priority, such as shutting down the space probe into a hibernation mode or restarting, cannot be interrupted. However, the tasks for scientific purposes which have lower priority and long duration can be interrupted. So it can be assigned to discrete relay windows assuming it meets the demand. According to whether the task can be interrupted by others, tasks can be divided into two types: strict satisfaction and best satisfaction. For strict satisfaction tasks, the duration of the continuous relay windows must not be less than the duration of the task. For best satisfaction tasks, the assigned relay windows can be discontinuous. Also, the total duration of it can be less than the duration of task and it satisfies the following constraints:

$$T_1 \cup T_2 = T \tag{4}$$

$$\forall t_i \in T_1,\ |R_i| \leq 1 \tag{5}$$

$$\forall t_j \in T_2, \sum_{k=1}^{m} \left(e_{jk} - b_{jk}\right) \leq d_j \tag{6}$$

The matrix $X = (x_i)_{|T|}$ indicates the scheduling information of tasks , $x_i = 1$ indicates that the task t_i is successfully assigned to relay window, and $x_i = 0$ indicates that no relay resources are allocated.

2) Optimization goal

Based on information such as the importance and urgency of tasks, profits of each task is considered, and the scheduling profits are prioritized at the same time, that is, the maximum average priority of the execution task under the constraint condition.

$$\max\left(\frac{\sum_{i=1}^{n} p_i x_i}{\sum_{i=1}^{n} x_i}\right) \tag{7}$$

3) Sliding time window and task merging

Probes can communicate with the control center through the relay satellite only when the relay satellite is in the TT&C arc. The task must be performed within the visible arc of the relay satellite. At the same time, platform maintenance tasks such as orbit maintenance of relay satellite also need to be assigned to a part of the relay window. Therefore, the visible relay window that can be allocated to the probe is discontinuous, as shown in Fig. 2(a).

The model allocates resources to non-preemptible task t_i ($t_i \epsilon T_1$) preferentially, assuming that the time window assigned to the task is $[b_i,e_i]$, sliding the time window $[b_i,e_i]$ from left to right, as shown in Fig.2 (a)(b), such that $b_i \geq c_{ib}$ and $e_i \leq c_{ie}$. If $\exists w_j \in W$, and $w_{je} - w_{jb} \geq d_i$, then w_j is an available window for the task t_i. In order to meet the centrality constraint, the first window $[w_{je}, w_{je} + d_i]$ that satisfies the demand in the sliding time window process is assigned to t_i. Its corresponding remaining window S_i is $W - [w_{je}, w_{je} + d_i]$, as shown in Fig.2.(d). Iterating S_i as the visual relay window for task t_{i+1}, allocates to subsequent tasks in the queue.

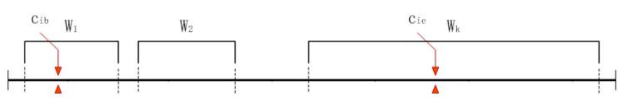

Fig.2. (a) Visual relay window

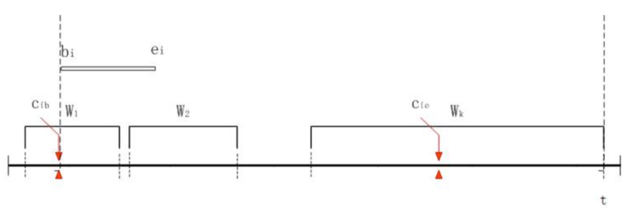

Fig.2. (b) Sliding window

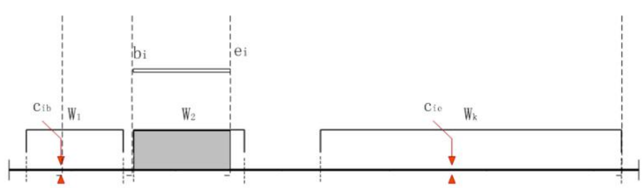

Fig.2. (c) Assigned window

Fig.2. (d) The remaining window

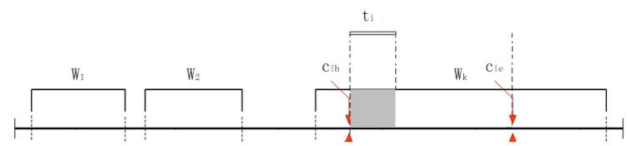

Fig.2. (e) Task conflict 1

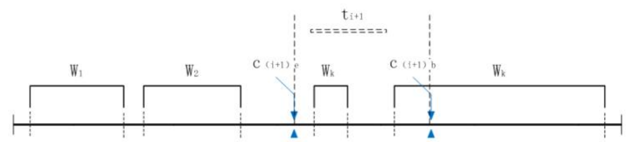

Fig.2. (f) Task conflict 2

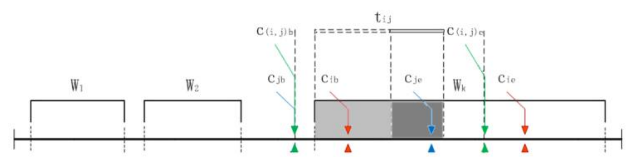

Fig.2. (g) Task merging

After allocating the relay window for t_i in the way described above, it might cause that the remaining relay windows do not meet the requirements of t_{i+1}, due to the cropping of the visual relay window, as shown in Fig2.(e)(f). In order to reduce the conflicts above-mentioned and the time fragmentation, this paper proposes a task merging strategy. If the constraint time window of tasks t_i and t_j satisfy $[c_{ib}, c_{ie}] \cap [c_{jb}, c_{je}] \neq \emptyset$, you can merge tasks t_i and t_j into a new task $t_{i,j}$, whose task duration is $d_i + d_j$, and the earliest start time $c_{(i,j)b}$ and latest end time $c_{(i,j)e}$ are:

$$C_{(i,j)b} = \begin{cases} \mathrm{Max}(c_{ib} - d_j, c_{jb}), & c_{ib} \geqslant c_{jb} \\ \mathrm{Max}(c_{jb} - d_i, c_{ib}), & c_{ib} < c_{jb} \end{cases} \tag{8}$$

$$C_{(i,j)e} = \begin{cases} \mathrm{Min}(c_{je} + d_i, c_i), & c_{ie} \geqslant c_{je} \\ \mathrm{Min}(c_{ie} + d_j, c_j), & c_{ie} < c_{je} \end{cases} \tag{9}$$

By using the sliding time window method, the available window $[r_{ijb}, r_{ije}]$ can be obtained from the merged task $t_{i,j}$, and it can be further decomposed into two windows $[r_{ijb}, r_{ijb}+d_j]$ and $[r_{ijb}+d_j, r_{ije}]$,which are the available relay window of task t_i and t_j, as shown in Fig2.(g).

V. Experimental Results

In the deep space celestial backside detection, the deep space relay resource planning model based on sliding time windows is applied. In practical application, according to the different task requirements of multiple probes, the 20-day relay resources are planned and allocated. Table 1 shows some examples of relay resources requirements. It shows various types of requirements and complex constraint relationships.

Table 1 User relay requirement (sample)

Detector	Priority	Length(min)	Mode	Fixed start time	Earliest start time	The latest end time
Relay satellite	53	120.00	0	XX-XXT04:50:00	——	——
Space probe B	51	100.00	0	XX-XXT16:50:00	——	——
Space probe A	0	650.00	1	XX-XXT03:00:00	XX-XXT02:00:00	——
Relay satellite	51	180.00	1	XX-XXT12:00:00	XX-XXT08:00:00	XX-XXT23:00:00
Space probe A	0	120.00	4	——	——	——
Relay satellite	11	280.00	4	——	——	——

In order to verity the applicability and feasibility of the model, 5 cases are constructed using actual data:

Table 2 Simulation Result

No	Quantity of visual relay window	Task quantity	Task completion rate
1	30	10	100%
2	30	20	95%
3	50	20	100%
4	50	30	97%
5	50	40	95%

The results of the relay resource planning are shown in Table 2. The model works well with 5 examples of different scales.

As a comparison, the FSS (First Sequence Serving) algorithm is designed. The relay resource planning model based on sliding time occupies 19 arcs, and the FSS model occupies 22 arcs. The effective utilization rate of relay resources in two models are 61.26% and 48.60%, as shown in Fig.3, Our model is increased by 12.66%. The resource fragmentation rate are 48.06% and 33.03%, As shown in Fig.4, and the resource fragmentation rate is reduced by 15.03%.

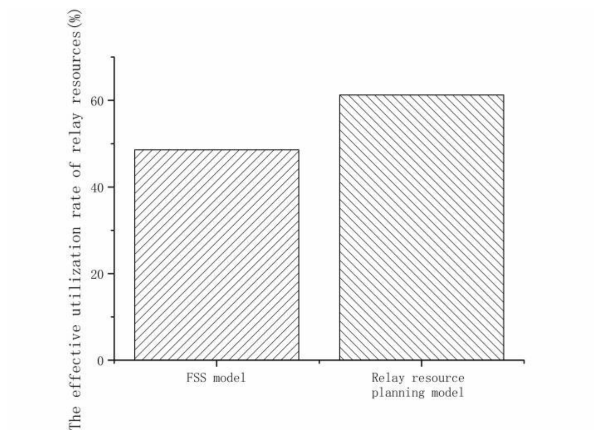

Fig.3. Effective utilization rate of relay resource

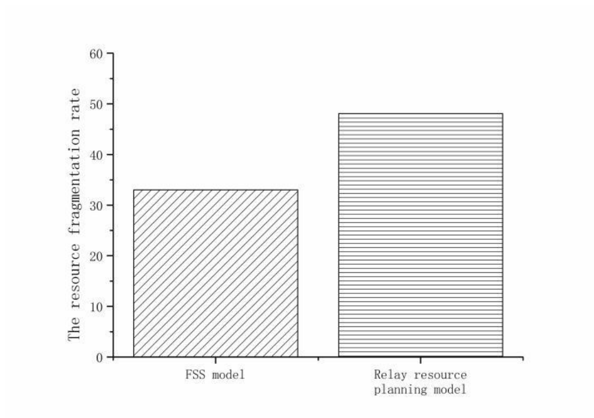

Fig.4. Resource fragmentation rate

VI. Conclusion

Aiming at the detection requirements of the back of deep space exploration, a system structure of hierarchical planning and scheduling of relay resources is designed. With study and analysis of the various constraints of detection tasks on relay resources, we established a relay resource planning model and designed the calculation method of the available time window for the task. In the limited arc segment, tasks with different user and different priorities are scheduled to execute. Through the application in the actual task of detecting the far side of celestial body, the relay resource planning algorithm based on sliding time can meet the resource allocation requirements of the tasks. It can improve task completion rate and reduce the fragmentation rate greatly. This model can effectively solve the contradiction between the supply and demand of deep space exploration relay resources. It provides theoretical foundation and technical support for the automatic planning of relay resources for subsequent deep space exploration missions, such as mars exploration missions.

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